COMSATS University Islamabad, Lahore Campus

Department of Electrical and Computer Engineering

Mid Term Examination – SPRING 2023

Course Title:	Machine Learning			Course Code:	CSC354	Credit Hours:	3(3,0)
Course Instructor:	Dr. Ikramullah Khosa			Program Name:	BS Computer Engineering		
Semester:	SP23	Batch:	FA19-BCE Section:	A,B	Date:	09-05-2023	
Time Allowed:	90 Minutes			Maximum Marks:		25	
Student's Name:	M Shaban Rana			Reg. No.	FA19 BCE- 014		

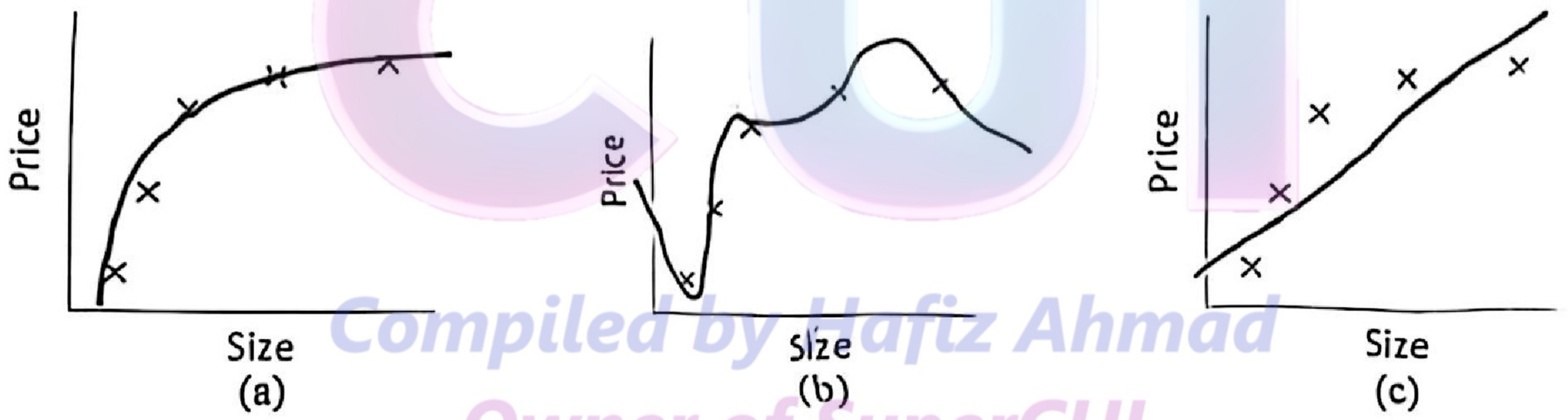
Important Instructions / Guidelines:

- This is a close-notes, close-books examination.

Question 1 (PLO2-CLO2-C4) **(10 Marks)**

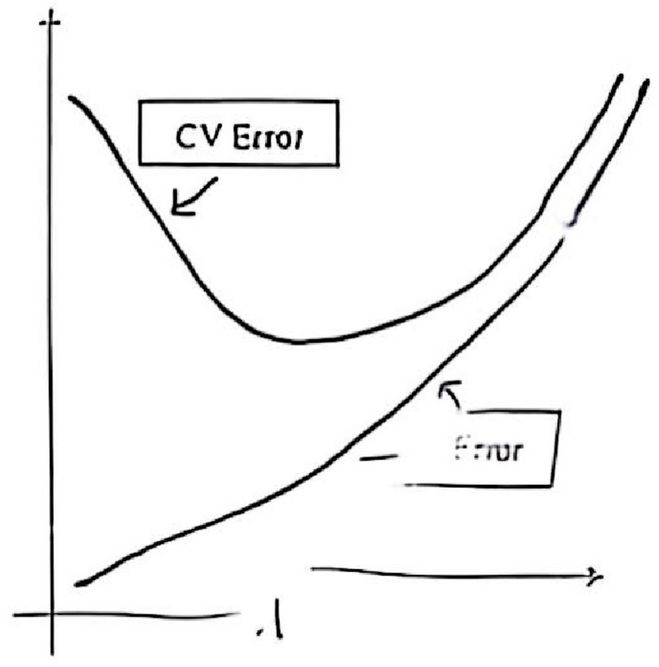
Relate the fundamental concepts of machine learning and answer the following short questions.

1) State two main types of machine learning. (2)
2) Assume a convolutional layer in a CNN model takes an input volume of 30×30×3. If 10 filters each of size 5×5×3 are applied without padding with a stride of 1. Analyze the computations at the convolution layer and estimate the output volume of the layer as well as the total parameters of the layer. (2)
3) Analyze the three cases in the following figure and identify the correct label for each as among: over fitting, under fitting and just right. (3)



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4) Analyze the training error and cross validation (CV) error curves against the regularization parameter (λ) in the following figure and identify the high bias, high variance regions, and just right regions. (3)



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Question 2 (PLO1-CLO1-C3) **(9 Marks)**

Consider the following data to train a linear regression classifier: where X = Three examples as input (one in each row), Y = corresponding outputs, Θ_1-Θ_3 = three model hypotheses. Compute the sum of absolute error $|X\Theta - Y|$ for each hypothesis and identify the best hypothesis model.

$$X = \begin{bmatrix} 1 & 10 \\ 1 & 21 \\ 1 & 14 \end{bmatrix}, Y = \begin{bmatrix} 200 \\ 270 \\ 223 \end{bmatrix}, \Theta_1 = \begin{bmatrix} 150 \\ 6 \end{bmatrix}, \Theta_2 = \begin{bmatrix} 195 \\ 1.5 \end{bmatrix}, \Theta_1 = \begin{bmatrix} 230 \\ 0.8 \end{bmatrix}$$

Question 3 (PLO1-CLO1-C3) **(6 Marks)**

Build and illustrate the computational graph for the following (sigmoid) function.

$$f(w,x) = \frac{1}{1+e^{-(w_0x_0+w_1x_1+w_2)}}$$

Perform the forward pass and backward pass. Compute and show the scores during the forward pass and the gradients during the backward pass. Use the following values for the variables $w_0=2$, $x_0=-1$, $w_1=-3$, $x_1=-2$, $w_2=-3$

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